

10-Day Intensive DSA Pre-Placement Training Agenda

Objective: To transition from basic syntax to advanced algorithmic problem-solving with a focus on industry-standard optimization and coding patterns.

Day 1: Computational Thinking & Complexity Analysis

- Focus: Building a "Performance-First" mindset.
- Core Topics:
 - Revision: Variables, Scoping, Memory Stack vs. Heap, and Function Overhead.
 - The Big O: Deep dive into Time ($O(n)$, $O(\log n)$, $O(2^n)$) and Space complexity.
 - Analytical Tools: Using Trace Tables for logic verification; identifying bottlenecks before coding.
 - Bit Manipulation Basics (XOR properties, checking even/odd, setting/clearing bits) — often used as "warm-up" interview questions.
- Industry Alignment: Justifying why an $O(n \log n)$ solution is preferred over $O(n^2)$ in production environments.
- Practice: Complexity evaluation of nested loops vs. recursive calls; Power of Two (Bitwise).

Day 2: Array Engineering & Linear Patterns

- Focus: Mastery of the most fundamental data structure.
- Core Topics:
 - Traversals: In-place swaps, rotation, and multi-pass algorithms.
 - Subarray Optimization: Kadane's Algorithm and Prefix Sums for range queries.
 - Difference Array technique for range updates.
- Industry Alignment: Emphasis on space-efficient (in-place) modifications to handle large datasets.
- Practice: Next Permutation, Best Time to Buy and Sell Stock, Product of Array Except Self.

Day 3: Hashing – The $O(1)$ Powerhouse

- Focus: Trading space for time.
- Core Topics:
 - Internal Mechanics: Hash functions, Collision resolution (Chaining vs. Open Addressing).
 - Patterns: Frequency mapping, "First Occurrence" tracking, and Pair-finding.
 - Hashing + Prefix Sum: Solving the "Subarray Sum Equals K" pattern.
- Industry Alignment: Critical for understanding Load Balancers, Caching (Redis), and DB Indexing.
- Practice: Two Sum, Group Anagrams, Longest Consecutive Sequence.

Day 4: String Processing & Manipulation

- Focus: Mastering immutable data and pattern matching.
- Core Topics:
 - Fundamentals: String pool, immutability, and efficient concatenation (StringBuilder/List).
 - Pattern Matching: Two-pointer approach to palindromes and reversals.
 - Sliding Window on Strings (Handling character frequency maps).

- Industry Alignment: Essential for log parsing, search engine tokenization, and data cleaning.
- Practice: Longest Common Prefix, String to Integer (atoi), Valid Anagram.

Day 5: Advanced Optimization (Two Pointers & Sliding Window)

- Focus: Converting $O(n^2)$ problems into $O(n)$.
- Core Topics:
 - Two Pointers: Opposite directional (Sorted arrays) vs. Same directional pointers.
 - Sliding Window: Fixed size (max sum of K elements) vs. Variable size (smallest subarray with sum S).
- Industry Alignment: Critical for real-time data streaming and network packet analysis.
- Practice: Container With Most Water, 3Sum, Minimum Size Subarray Sum.

Day 6: Recursion & Backtracking (The Decision Space)

- Focus: Exhaustive search and combinatorial optimization.
- Core Topics:
 - The Recursive Mindset: Identifying the "Self-Similar" subproblem.
 - Backtracking: "Choose, Explore, Un-choose" template.
 - Pruning: Cutting the search tree early to save time.
- Industry Alignment: Used in AI pathfinding, circuit design, and scheduling constraints.
- Practice: Permutations, Sudoku Solver, Word Search.

Day 7: Memory Management (Linked Lists) & Monotonic Linear Structures

- Focus: Physical vs. Logical data ordering.
- Core Topics:
 - Linked Lists: Pointer manipulation, Floyd's Cycle Detection (Tortoise & Hare).
 - Stacks/Queues: LIFO/FIFO applications.
 - Added Missing: Monotonic Stack/Queue (Finding the Next Greater Element)—a very frequent "Hard" interview pattern.
- Industry Alignment: Low-level system design (LRU Cache) and OS task scheduling.
- Practice: LRU Cache (Medium/Hard), Reverse Linked List in K-Groups, Daily Temperatures.

Day 8: Hierarchical Structures (Trees)

- Focus: Navigating non-linear relationships.
- Core Topics:
 - Traversals: Deep dive into DFS (Recursive) vs. BFS (Iterative using Queue).
 - Binary Search Tree (BST) Logic: Insertion, deletion, and finding the K-th smallest element.
 - Tree Construction: Building a tree from Inorder/Preorder traversals.
- Industry Alignment: DOM Tree (Web Browsers), Folder Structures, and JSON Parsing.
- Practice: Serialize and Deserialize Binary Tree, Binary Tree Zigzag Traversal, Validate BST.

Day 9: Graphs & Greedy Heuristics

- Focus: Network modeling and greedy decision-making.
- Core Topics:

- Graph Basics: Adjacency List/Matrix, DFS/BFS on Graphs.
 - Topological Sort: Handling dependencies (Kahn's Algorithm).
 - Greedy: Interval scheduling and sorting-based greed.
 - Added Missing: Dijkstra's Algorithm (Basics of shortest path).
- Industry Alignment: Recommendation engines, GPS (Google Maps), and Build Systems (Maven/Gradle).
- Practice: Course Schedule, Number of Provinces, Gas Station (Greedy).

Day 10: Dynamic Programming & Placement Simulation

- Focus: Ultimate optimization and "Thinking under pressure."
- Core Topics:
 - DP Fundamentals: Identifying "Overlapping Subproblems" and "Optimal Substructure."
 - Patterns: 0/1 Knapsack, Longest Common Subsequence, Grid-based DP.
 - Mock Simulation: 60-minute timed challenge followed by code review.
- Industry Alignment: Financial modeling, resource allocation, and sequence alignment in biotech.
- Practice: Climbing Stairs, Coin Change, Longest Increasing Subsequence.



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